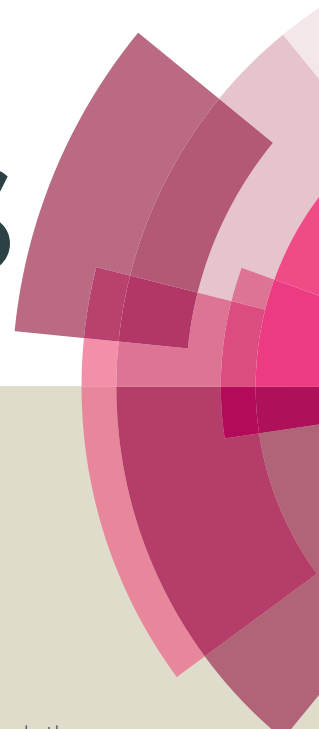


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COMMUNICATION

Synthesis of allyl-aziridines from α -Halo Oxime Ethers and Allyl zinc bromides

Received 00th January 20xx,
Accepted 00th January 20xxSonglin Zheng and Songlin Zhang^{*a}

DOI: 10.1039/x0xx00000x

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A novel method for the preparation of substituted allyl aziridines by reaction of α -halo with allylic zinc reagents in mild conditions is reported in this paper. The present method complements the existing synthetic methods due to some advantages offered by the use of organozinc reagents, which are easily prepared, relatively stable and non-toxic, and more selective than Grignard reagents.

The aziridine ring, the smallest nitrogen heterocycle, is present in the structure of a wide variety of natural products which exhibit various biological properties. For example: compound A (ethylenimine quinone)¹ (Figure 1) is a synthetic drug with antileukemic effect; B (mitomycin)² and C (azinomycin)³ are natural products with anticancer effect. In addition, aziridines are useful building blocks for the synthesis of organonitrogen compounds due to easy opening and expansion of the three-membered ring.^{4a}

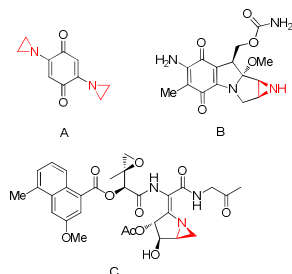


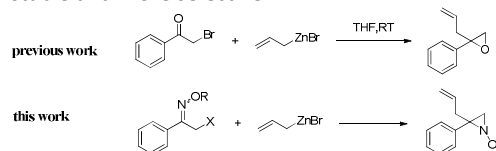
Figure 1 some aziridine drugs.

Owing to these reasons, the synthesis of aziridines has great significance in organic chemistry and there have been a large number of reports pertaining to the synthesis of aziridines⁴ since the first synthesis reported by Gabriel⁵ in 1888. Wenker synthesis⁶ and aza-Darzens approaches⁷ are the classical synthetic methods. However, strong base or acid must be used

in these reactions. Besides, aziridination of alkenes with nitrenes is an important approach to aziridines. Nitrenes are obtained by oxidation of nitrogen compounds⁸ or the decomposition of azides⁹. The range of oxidant and amines allowed are very narrow and a sulfonyl group is generally introduced into the target compounds. The usage of oxidant or azide is usually dangerous even operating in laboratory scales. Transfer of carbene to imines is an alternative method to synthesize aziridines, but low temperature¹⁰ or explosively-dangerous diazo compounds¹¹ are needed in this type of reaction. Grignard reagents with oximes^{12a, 12b} or α -halo imines^{12c, 12d} is a safer method to construct aziridine compounds although Grignard reagents are not compatible with many functional groups.

In view of the importance of this class of compound and the limitation of previous synthetic methods, effective procedure for preparing aziridines is desirable.

Organozinc reagents are among the most useful organometallic reagents and have been widely used in organic and organometallic synthesis.¹³ Compared with Grignard reagents, organozinc reagents are easy to prepare, relatively stable and more selective.¹⁴



Scheme 1 previous work and this work.

Researches on organozinc reagents have been performed in our laboratory over the past few years.¹⁵ Among these works, we have developed an efficient synthetic method for the preparation of allyl-epoxides by allylation of α -haloketones with organozinc reagents (scheme 1).^{15a, 15c} Based on previous work, the exploration of a new protocol for the synthesis of allyl-aziridines via allylation of α -bromo oxime ethers with allylzinc bromides is presented in this communication.

In our initial study, 2-bromo-1-phenylethanone *O*-methyl oxime (**1a**, *E/Z* mixture) was chosen as a model substrate to

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Electronic Supplementary Information (ESI) available: [details of any supplementary information available should be included here]. See DOI: 10.1039/x0xx00000x

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react with allylzinc bromide (**2a**) in THF at room temperature and after 14 hours the corresponding aziridine **3a** was obtained in 85% yield. Encouraged by this result, the scope of the reaction of α -bromo oxime ethers with allyl- and methallylzinc bromides was examined. The results are summarized in the Table.

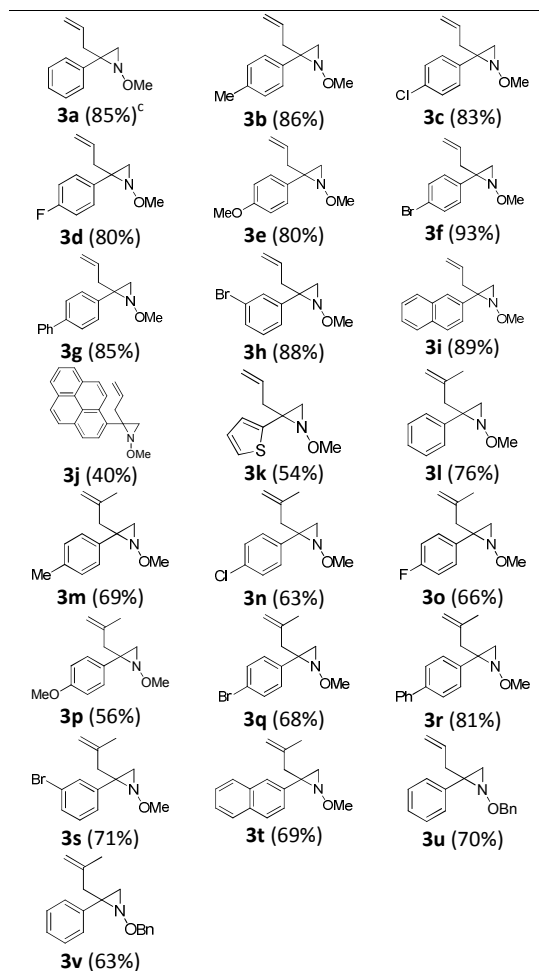
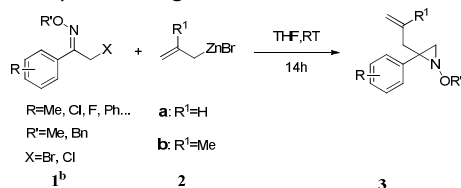
We primarily investigated the reaction of allylzinc bromide **2a** with a series of 1-aryl-2-bromoethanone *O*-methyl oximes differing for the nature and position of the substituent of the phenyl ring. The results reported in the Table 1 indicated that both electron-withdrawing and -donating groups were tolerated, and the products **3a-h** were obtained in good yields (80-93%). Moreover, When 2-bromo-1-(naphthalen-2-yl) ethanone *O*-methyl oxime was used as substrate, the corresponding product **3i** could be obtained in a yield of 89%. However, 2-bromo-1-(pyren-1-yl)ethanone *O*-methyl oxime(**1j**) was less reactive and provided the product **3j** with only 40% yield. Also, the allylation of a heterocyclic substrate such as 2-bromo-1-(thiophen-2-yl) ethanone *O*-methyl oxime (**1k**) gave the corresponding aziridine **3k** in 54% yield.

The reactions of the α -bromo oximes with methallylzinc bromide **2b** were examined and a decrease in the reactivity became evident (Table 1). As a matter of fact, the aziridines **3l-3t** were obtained in satisfactory yields, although lower than those obtained using **2a**, compare for example compounds **3d/3o**, **3e/3p**, **3f/3q**, **3g/3r** and **3i/3t**. The highest yield was obtained for compound **3r** (81%). The lower yields may be due to the larger steric hindrance of the 2-methylallyl group.

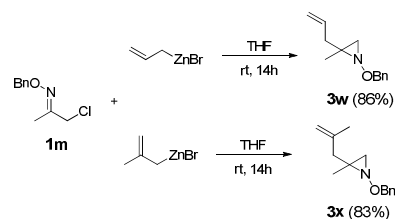
Then, we performed the reaction of allylic zinc bromide (**2a** and **2b**) with 2-bromo-1-phenylethanone *O*-benzyl oxime (**1l**). It was obvious that **1l** can give the corresponding products in lower yields than **1a** (Table 1, **3a/3u**, **3l/3v**). The reactions of 1-chloropropan-2-one *O*-benzyl oxime (**1m**) with **2a** and **2b** were also examined and the corresponding aziridines were obtained in good yields (Scheme 2).

In order to test the compatibility of allylzinc bromide with cyano and ester functions, we examined the reaction of allylzinc bromide **2a** with 4-(2-bromo-1-(methoxyimino)ethyl) benzonitrile (**1n**) and methyl 4-(2-bromo-1-(methoxyimino)ethyl)benzoate (**1o**). Unfortunately, the reaction of **1n** with allylzinc bromide **2a** leads to a complex mixtures and no corresponding product was detected (Scheme 3). It can be presumed that the cyano group is attacked first by **2a**. However, when the zinc reagent **2a** was used the preliminary attack to the ester group was observed, and the imino group was attacked later, so that the full allylation product **3y** (Scheme 3) was obtained in 60% yield only after addition of 6 equivalents of **2a**.

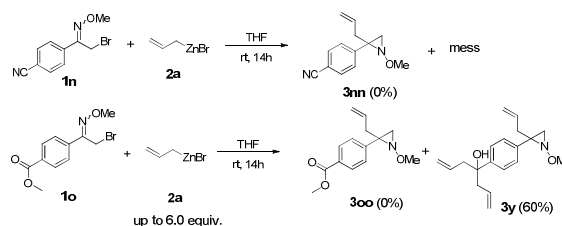
Table 1 Synthesis of allyl-aziridines from alpha-halo-oxime ethers and allylic zinc reagents^a



^a Reaction conditions: α -bromo oxime (0.5 mmol), allylzinc bromide (1 mmol) and THF (4mL) at room temperature under nitrogen for 14 h. ^b E/Z mixture. ^c Isolated yield.



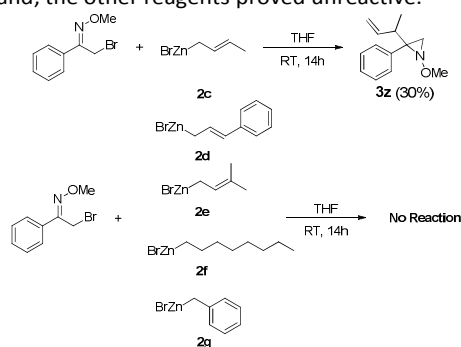
Scheme 2 Reactions of **1m** with **2a** and **2b**



Scheme 3 Reactions of **2a** with **1n** and **1o**

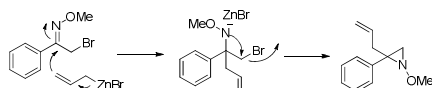
The reactions of the prototypical oxime ether with other substituted allyl-, benzyl- and octylzinc bromides were investigated (Scheme 4). But-2-en-1-ylzinc bromide (**2c**) gave 2-(but-3-en-2-yl)-1-methoxy-2-phenylaziridine (**3z**) although in

low yield as a mixture of diastereoisomers (d.r. > 20:1). On the other hand, the other reagents proved unreactive.



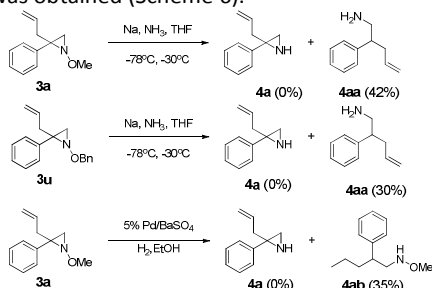
Scheme 4 Reactions of **1a** with other organozinc reagents

According to the results above, a possible two-steps reaction pathway is deduced in Scheme 5.



Scheme 5 The possible reaction pathway

At last, some efforts have been made to remove the OMe group on aziridine according to the literatures^{4j, 16}; unfortunately, no corresponding aziridine with an OMe group cleaved was obtained (Scheme 6).



Scheme 6 attempts to remove the N-alkoxy substituent from aziridines

In conclusion, series of substituted 2, 2-disubstituted-N-alkoxy aziridines in good yields by allylation of α -halo oxime ethers with allylic zinc reagents. The reaction proved to be tolerant to a variety of substrates, and could be carried out in a one-pot manner at ambient temperature. These features make the current method more attractive to application in industrial scale.

We would like to gratefully acknowledge financial support from the A Project Funded by the Priority Academic Program for the Development of Jiangsu Higher Education Institutions, The Project of Scientific and Technologic Infrastructure of Suzhou (No. SZS201207) and the National Natural Science Foundation of China (No. 21072143).

Notes and references

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Electronic supplementary information (ESI) available: Experimental procedures, characterization and spectral data of the products. See DOI: 10.1039/b000000x/

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